

Abubakar Sadiq Shittu

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Education

Ph.D. Computer Science, *University of Tennessee, Knoxville*

Jan 2025 – Anticipated: Dec 2027

GPA: 3.67/4.00 | Coursework: Advanced Software Engineering, Algorithms, Data-Driven Security & Privacy, Computer Systems Organization, Digital Archaeology.

M.Sc. Advanced Computer Science, *Newcastle University, United Kingdom*

Sept 2023 – Sept 2024

GPA: 4.00/4.00 (Distinction) | Coursework: Machine Learning, Statistical Modelling, Secure Software Development, Systems Security, Risk & Trust Management.

B.Sc. Computer Science, *Fountain University, Osogbo*

March 2018 – June 2021

GPA: 4.63/5.00 (First Class Honours)

Experience

Graduate Research Assistant

Jan 2025 – Present

Usable Security and Engineering Research Laboratory, Led by **Dr. Scott Ruoti**

United States

- Engineered a high-throughput, containerized data ingestion pipeline using Docker and Celery for distributed task scheduling; integrated GitHub REST and GraphQL APIs with a PostgreSQL backend to continuously harvest 4M+ user profiles and 20M+ commit records; surfaced adoption trends and behavioural patterns across the ecosystem [1].
- Spearheaded longitudinal mixed-methods studies on (1) TLS client authentication, triangulating participant feedback, server logs, and 400+ certificate signing requests to identify usability failures [2]; and (2) git commit signing, validating qualitative findings against .git log data and usability benchmarks (SUS/ASQ) to pinpoint technical and usability barriers to its adoption as a supply chain defense [manuscript in progress].
- Conducting an interview study with 20+ software developers, synthesizing qualitative mental-model data with prior large-scale quantitative findings to explain adoption barriers [manuscript in progress].

Information Technology Helpdesk Intern

Mar. 2019 – Aug. 2019

Nigerian National Petroleum Corporation

Nigeria

- Supported 100+ users as primary technical contact, resolving 5+ daily issues and maintaining operational continuity.

Selected Class Projects

Machine Learning Pipelines, Newcastle University, UK

Dec 2023

- Built end-to-end pipelines for classification, regression, and NLP clustering across four tasks: (1) tuned Random Forest on FARS (100K+ records), benchmarking SVM, Gradient Boosting, and Logistic Regression with SMOTE; (2) predicted bacterial growth parameters (μ , τ , a_0) across 19,992 experiments; (3) compared TF-IDF vs. embeddings for K-Means clustering on 2,225 news articles; and (4) tuned an LSTM on Raspberry Pi sensor data; produced visualizations and performance plots using Matplotlib throughout.

Advanced RAG Pipeline

Spring 2025

- Built a question-answering pipeline using LangChain and Gemini with RecursiveCharacterTextSplitter chunking and VertexAIEmbeddings for retrieval; scraped a university course schedule with BeautifulSoup, cleaned and stored it as JSON, and indexed it in a vector store to answer natural-language queries.

LLM-Based Assistant for Dog Care Guidance

Spring 2025

- Curated a veterinary-verified dataset of 260 dog breeds using Python and BeautifulSoup; used Gemini via LangChain to extract structured JSON data for each breed.
- Built and compared two systems: a RAG pipeline (TinyLlama-1.1B + FAISS) and a LoRA fine-tuned LLaMA-3.2-1B model; RAG outperformed fine-tuning across 15 expert-verified dog care questions ($p < 0.05$).

Preprints

- [1] **A. S. Shittu**, J. Sadik, F. Gholamrezae, and S. Ruoti, “Analysis of commit signing on GitHub,” *arXiv preprint arXiv:2604.14014*, 2026. [Online]. Available: <https://arxiv.org/pdf/2604.14014>.
- [2] **A. S. Shittu**, C. Shubert, J. Sadik, and S. Ruoti, “Understanding student experiences with TLS client authentication,” *arXiv preprint arXiv:2604.14330*, 2026. [Online]. Available: <https://arxiv.org/pdf/2604.14330>.

Scholarships and Awards

- **Travel Grant**, *Summer School on Real-World Crypto and Privacy, Vodice, Croatia* 2024
- **Chevening Scholarship**, *United Kingdom*. ~\$70,000: tuition, monthly stipend, health insurance, and transportation 2023

Skills, Service, and Affiliations

- **Development:** Python (pandas, NumPy, scikit-learn, Matplotlib, statsmodels), R, C/C++, Docker, SQL, PostgreSQL, REST/GraphQL APIs | **Systems & Analysis:** macOS, Windows, Linux/Unix, Bash, SonarQube, Wireshark, Git, NVivo, MAXQDA.
- **Reviewing:** [Journal of Cybersecurity](#), Q1 (SJR), 2025 | **Affiliations:** Graduate Student Member, IEEE & IEEE Computer Society | **Volunteering:** [Tennessee Extension Volunteer](#)